TATA CLiQ E-COMMERCE DATABASE MANAGEMENT SYSTEM

SQL QUERY IMPLEMENTATION

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table. Display all products:

```
SQL> SELECT * FROM PRODUCT;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
       101
Samsung Galaxy S24
Latest Samsung smartphone
     69999         15           2
```

1.2

WHERE Clause

The WHERE clause is used to filter records based on a specified condition.

Find product ID and name where category ID is 1:

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
DESCRIPTION
--------------------------------------------------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
       106
Samsung Smart TV
Smart LED television
     45999          8           1
```

```
SQL> SELECT *
  2  FROM PRODUCT
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
----
DESCRIPTION
--------------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
       101
Samsung Galaxy S24
Latest Samsung smartphone
     69999         15           2
```

1.3 ORDER BY

The ORDER BY clause is used to sort query results in ascending or descending order.

Display products in descending order of price:

```
SQL> SELECT *
  2  FROM PRODUCT
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
----
DESCRIPTION
--------------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
       101
Samsung Galaxy S24
Latest Samsung smartphone
     69999         15           2
```

Display products in ascending order of price:

```
SQL> SELECT *
  2  FROM PRODUCT
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
----
DESCRIPTION
--------------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
       204
Grocery
Grocery Items
       500         30          4
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values and eliminate duplicate entries.

Display all unique category IDs:

```
SQL> SELECT DISTINCT CATEGORY_ID
  2  FROM PRODUCT;

CATEGORY_ID
-----------
          2
          3
          5
          6
          1
          8
          9
          4

8 rows selected.
```

2. Search Products

2.1 Search Products Based on Price

Products can be searched using their price to identify items within a particular price range.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 1000 AND 25000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
----
     PRICE
----------
       104
Nike Running Shoes
      3999

       107
Fastrack Watch
      1299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
----
     PRICE
----------
```

2.2 Search Products Based on Category

Products can be filtered according to their category**.**

Display products belonging to category ID 1:

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------
----
DESCRIPTION
--------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
       106
Samsung Smart TV
Smart LED television
     45999          8           1
```

3. Retrieve Customer Information

The SELECT statement is used to retrieve customer details from the Customer table.

Display all customers:

```
SQL> SELECT * FROM CUSTOMER;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
----
PHONE
---------------
PASSWORD
--------------------------------------------------------------------------------
----
ADDRESS
--------------------------------------------------------------------------------
----
          1 Thansila

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
----
PHONE
---------------
PASSWORD
--------------------------------------------------------------------------------
----
ADDRESS
--------------------------------------------------------------------------------
----
Begam

CUSTOMER_ID FIRST_NAME
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT operators to filter records.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE < 1000
  4  AND STOCK > 0;

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
----
DESCRIPTION
----------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
       105
Lakme Face Cream
Moisturizing face cream
       899         30          6
```

Display products belonging to category 1 or category 2:

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID IN (1, 2);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
       101
Samsung Galaxy S24
Latest Samsung smartphone
     69999         15           2
```

Display products that do not belong to category 1:

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID <> 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
----
DESCRIPTION
--------------------------------------------------
----
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
       101
Samsung Galaxy S24
Latest Samsung smartphone
     69999         15           2
```

5. Basic Business Reports

Business reports help analyse product prices, customer orders, and product sales.

5.1 Product Price Report

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  ORDER BY PRICE DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
     PRICE
----------
       101
Samsung Galaxy S24
     69999

       102
iPhone 15
     64999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
     PRICE
----------

       103
```

5.2 Customer-wise Total Orders

This report calculates the total number of orders placed by each customer.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- ------------
          1 Thansila
Begam                                                         0

          2 Priya
Devi                                                          0

          4 Divya
Rani                                                          0
```

5.3 Customer-wise Total Purchase Amount

This report calculates the total purchase amount for each customer based on their orders.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         NVL(SUM(O.TOTAL_AMOUNT), 0) AS TOTAL_PURCHASE_AMOUNT
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
          1 Thansila
Begam                                                                  0

          2 Priya
Devi                                                                   0

          4 Divya
Rani                                                                   0


CUSTOMER_ID FIRST_NAME
```